
Computational Models for Polycystic Ovary Syndrome: An Endocrinological Approach to Dysregulation

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Abstract

Polycystic ovary syndrome (PCOS) impacts millions of women in the United States alone, yet is often left undiagnosed. This difficulty in diagnosis can be attributed, in part, due to the nature of its symptoms and etiology—both of which are multifaceted. Current computational work concerning PCOS has focused on multimodal data, such as genetic predispositions, or metabolic markers. However, there is a lack of computational research from a hormonal framework. In this paper, I will focus on two models of dysregulated hormone feedback loops: 1) the increase of LH to FSH ratio, and 2) insulin resistance. I will also discuss a unified, systemic approach: modeling dysregulation of the hypothalamic-pituitary-gonadal (HPG) axis. Lastly, machine learning approaches bring insight towards potential risk factors and significant predictors. Through both mechanistic and data-driven approaches, this review moves towards more robust, clinical applications.

1 Introduction

Polycystic ovary syndrome (PCOS) is a common endocrine disorder that impacts reproductive-aged women, yet studies on its etiology are relatively limited. An estimated 6 to 13 percent of the susceptible population have PCOS, making it the most common endocrine disorder among reproductive-aged women [2, 38]. PCOS has a broad physiological and mental impact on women, including reduced fertility, fatigue, physical limitations, and other lifestyle-related factors. As evidenced by the eight domains of SF-36 (the standardized health survey to analyze a patient's health-related quality of life), physical, social and emotional well-being is heavily affected [46]. In addition to its well-known effects on fertility, PCOS also increases the risk of other chronic health conditions. More than half of the women with PCOS develop type 2 diabetes by the age of 40 [1].

However, an estimated 75 percent of people with PCOS remain undiagnosed. Although symptoms can develop as young as 11 or 12, women are generally not diagnosed until they try to become pregnant. This low diagnostic capability is partly due to the heterogeneous, broad physiological symptoms of the disorder: most commonly acne, hair growth, weight gain, darkening of the skin in the body creases, and the lack of oocyte release from the ovaries [38]. These symptoms overlap with those of other endocrine disorders, such as endometriosis (where there is abnormal tissue growth outside the uterus), further presenting challenges to diagnosis [20].

Multiple diagnostic criteria for PCOS have been proposed, but the most widely accepted is the 2003 Rotterdam criteria. According to the Rotterdam criteria, three features must be considered: 1) oligo-amenorrhea (irregular or absent menstrual cycles), 2) hyperandrogenism (elevated levels of androgens throughout the body, such as testosterone), and 3) polycystic-appearing ovarian morphology upon ultrasonography (presence of follicles—fluid-filled sacs that are underdeveloped, immature eggs, often mistaken for cysts—in the ovaries) [39]. In this review, I will utilize the Rotterdam criteria to diagnose PCOS. Given the three features proposed by Rotterdam, PCOS has three four major clinical phenotypes: phenotypes A, B, C, and D, ordered from most to least severe. Phenotype A exhibits hyperandrogenism, anovulation, and polycystic ovaries. Phenotype B exhibits hyperandrogenism and anovulation, but not polycystic ovaries. Phenotype C exhibits hyperandrogenism and polycystic ovaries, but not anovulation. Lastly, phenotype D exhibits anovulation and polycystic ovaries, but not hyperandrogenism (Table 1).

	Phenotypes			
	A	B	C	D
Symptoms				
Hyperandrogenism	X	X	X	
Anovulation	X	X		X
Polycystic Ovaries	X		X	X

Table 1: Four major PCOS phenotypes according to the Rotterdam criteria, ordered from most (A) to least (D) severe

Although the 2003 Rotterdam criteria is of relevance for diagnosis in clinical settings, recent studies have suggested that revisions are necessary. Of these revisions, there should be a large focus on the hormonal imbalances that contribute to PCOS. This is because clinical symptoms correlated with PCOS, such as obesity, are not necessary symptoms in PCOS patients—which again, furthers diagnosis challenges. Both obese and non-obese PCOS patients, however, still exhibit similar imbalances in hormones: increased LH to FSH ratio, insulin resistance, prolactin, or thyroid-stimulating hormone (TSH) [41]. Moreover, features like insulin resistance are present in 35% to 85% of PCOS patients [3]. Indeed, Azziz et. al propose that, based on a review of medical literature, PCOS should primarily be considered as a disorder of androgen excess. This review focuses on PCOS as a disorder of: 1) insulin resistance, 2) increased LH to FSH ratio, and 3) androgen excess in the form of hypothalamic-pituitary-gonadal (HPG) axis dysregulation. This framework is limited, though, in that there are more hormonal imbalances that contribute to the manifested symptoms of PCOS.

2 Biological Context to PCOS

2.1 Relevant Hormones

2.1.1 Gonadotropin-releasing-hormone (GnRH)

Gonadotropin-releasing hormone (GnRH) is vital to the hypothalamic-pituitary-gonadal (HPG) axis, which regulates the production and release of sex hormones [10]. As such, under typical conditions, GnRH acts as a neuromodulator of sexual behavior. It regulates puberty onset, sexual development, and ovulatory cycles

in females [10]. However, in patients with PCOS, high-frequency GnRH changes can contribute to ovarian hyperandrogenemia. This is because frequency changes contribute to a signaling cascade: higher frequencies preferentially increase luteinizing hormone secretion [30] (see Figure 1).

2.1.2 Luteinizing hormone (LH)

In both sexes, LH is produced in the gonads and is a key player in maturation of primordial germ cells. In women, LH helps regulate the length and order of the menstrual cycle. With regards to the signaling cascade from high-frequency GnRH changes, an increase in LH secretion stimulates the theca cells in the ovaries, which increases androgen production [33]. LH binds to G-protein–coupled receptors on theca cells, triggering the activation of adenylyl cyclase. This enzyme then generates cyclic AMP (cAMP), leading to an increase in its intracellular levels. The rise in cAMP activates protein kinase A (PKA), which phosphorylates specific intracellular proteins. Finally, these proteins then mediate LH’s physiological effects, such as steroid production and ovulation [33].

2.1.3 Follicle-stimulating hormone (FSH)

Follicle-stimulating hormone (FSH) is co-produced with LH in the gonads. In females, follicle-stimulating-hormone (FSH) chiefly regulates estrogen production and follicular development. It plays a central role during the menstrual cycle, where FSH stimulates the maturation of ovarian follicles. Then, FSH secretion is suppressed as a dominant follicle begins secreting estradiol and inhibin. Once a threshold of estradiol levels—200 to 300 pg/mL for 48 hours— is crossed, the hypothalamus rapidly secretes GnRH, causing FSH and LH to peak and begin ovulation [35]. Subsequently, lowered FSH levels in people with PCOS have impaired conversion to estrogens.

2.1.4 Insulin

Insulin, a hormone released by the pancreas, is typically used for glucose metabolism. By helping cells absorb glucose from the bloodstream, insulin can lower blood sugar levels. It can trigger the conversion of glucose to glycogen and excess glucose to triglycerides. Insulin resistance is a condition in which the body’s insulin sensitivity decreases, thereby causing high blood sugar and increased adipose tissue. Prior research by Purwar et. al even suggest that insulin resistance may be the most significant pathophysiological trait in PCOS patients. Purwar et. al focus on PCOS as a disease partially caused by chronic low-grade inflammation; insulin resistance increases blood sugar levels, triggering oxidative stress and cytokine production through macrophages in adipose tissue [37] (Figure 3).

2.1.5 Testosterone

Although known as the primary male sex hormone, testosterone has regulatory roles in females as well. In females, testosterone can maintain bone density and regulate follicle development, as well as estrogen production. When in its dysregulated form—i.e. at excess levels, as in PCOS—testosterone can be responsible for excess hair growth in abnormal regions, anovulation, and alopecia [27].

2.2 Hypothalamic-Pituitary-Gonadal (HPG) Axis

As a major signaling pathway, the hypothalamic-pituitary-gonadal (HPG) axis regulates the reproductive system and sexual development in both males and females. As a result, malfunction of the axis is correlated with reproductive disorders, such as PCOS, hypergonadism, or hypogonadism [31]. The HPG axis is composed of three parts: 1) the hypothalamus, 2) the pituitary gland, and 3) the thyroid gland. Altogether, these parts participate in a negative feedback loop. The hypothalamus releases GnRH, and GnRH travels to the anterior pituitary where it acts on GnRHR. Next, signaling from GnRHR causes the secretion of gonadotropins, such as LH and FSH. LH then acts on its receptor in the gonads, stimulating the release of sex steroids, androgens, and estrogens. Lastly, these sex steroids inhibit the release of GnRH, completing a negative feedback loop and regulating hormone secretion [8] (Figure 1).

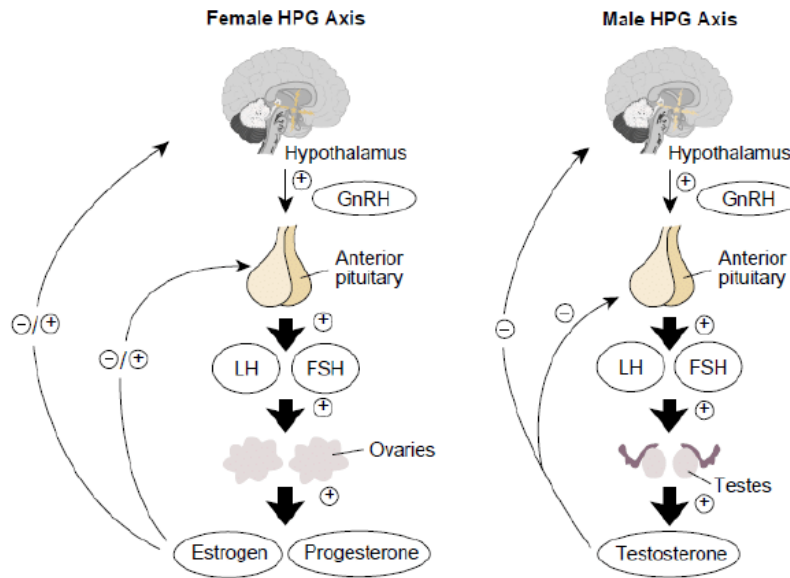


Figure 1: Hypothalamic-pituitary-gonadal (HPG) axis in females and males. The figure displays the secretion of GnRH from the hypothalamus, the downstream release of LH and FSH from the anterior pituitary, and the resulting production of sex hormones (estrogen/progesterone in females, testosterone in males). Negative (-) and positive (+) feedback loops are indicated [26].

2.3 Etiology of PCOS

PCOS patients exhibit an increased LH:FSH ratio and decreased FSH secretion. In addition, hyperandrogenism refers to elevated levels of androgens—particularly testosterone—which may result from increased androgen production and/or decreased levels of sex hormone-binding globulin (SHBG). SHBG is a protein produced by the liver that binds tightly to sex hormones like testosterone. Consequently, when SHBG levels are low, more free (active) testosterone circulates in the bloodstream, intensifying the symptoms of hyperandrogenism. Finally, patients with increased insulin levels and decreased glucose tolerance are said to have insulin resistance. It is important to note that these dysregulations may be triggered by a multitude of etiologies, many of which

have yet to be discovered. As of now, various causes include environmental pollutants, diet, lifestyle, genetics, and gut microbiota [44] (Figure 2).

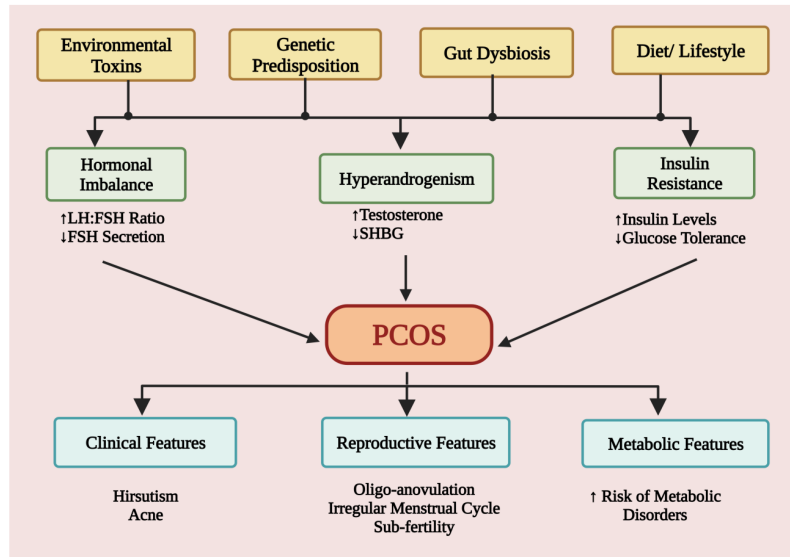


Figure 2: Overview of etiological factors contributing to PCOS, including environmental toxins, genetic predisposition, gut dysbiosis, and diet/lifestyle factors. These factors lead to hormonal imbalances, hyperandrogenism, and insulin resistance, ultimately manifesting in clinical, reproductive, and metabolic features [44].

3 LH and FSH Modeling

3.1 Sequential Monte Carlo

3.1.1 Pulsatile Hormone Secretion

Due to the homeostatic nature of the endocrine system, glands communicate with their remote target cells via both continuous and intermittent signal exchange. These exchanges are also known as basal and pulsatile dynamics, respectively. In basal secretion, endocrine glands maintain a steady-state concentration, so hormones are released at low levels. Meanwhile, pulsatile secretion is responsible for physiological responses. Thus, adjustments to hormone concentrations are made in quick, more extreme “pulses” [49]. GnRH, which regulates stimulation of the pituitary gland, is released in such a pulsatile manner. The pituitary gland is responsible for release of both LH and FSH, which makes GnRH an essential hormone to investigate in PCOS [49]. A hallmark in PCOS patients is elevated GnRH/LH pulsatility, but pulsatility is often resource intensive to assess. This is because LH must be measured in frequent, 10-minute intervals. A model of pulsatile secretion aligns with current physiological understanding of the endocrine system, and is strongly supported by the hypothalamic pulse generator hypothesis synthesized from murine models [11].

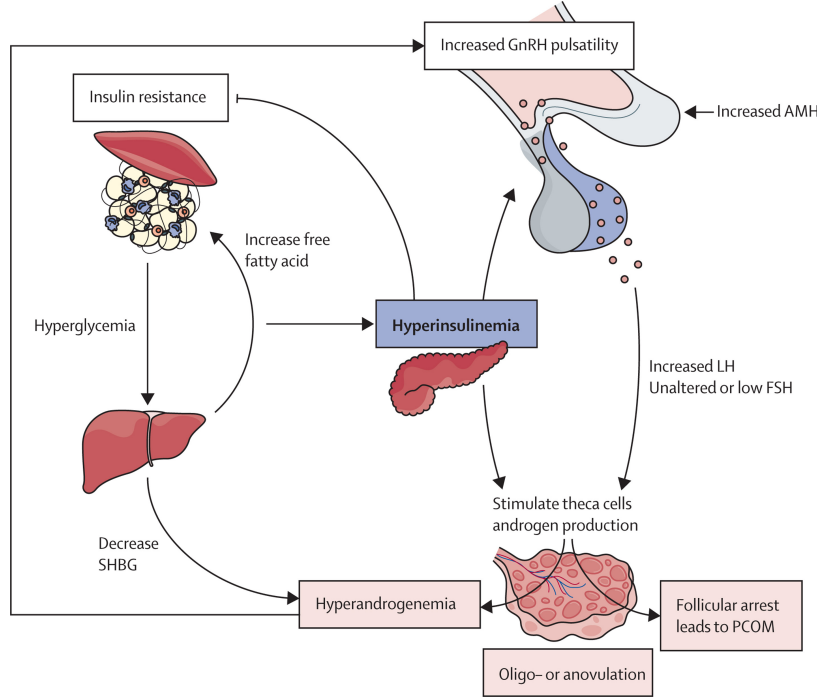


Figure 3: Illustration of how insulin resistance contributes to PCOS. Insulin resistance leads to hyperinsulinemia, which lowers SHBG levels and increases androgen production via stimulation of theca cells. These hormonal changes contribute to the central features of PCOS pathophysiology [25].

3.2 hormoneBayes Model

hormoneBayes analyzes GnRH/LH activity in PCOS patients via both pulsatile and basal dynamics. Pulsatile secretion is treated as an on/off signal that switches between high and low activity states, whereas basal secretion is continuously noisy. The study predicts LH circulating dynamics in terms of five key parameters that can be recovered from data: 1) LH clearance rate, 2) maximum LH release rate, 3) strength of the pulsatile signal relative to the basal signal, 4) mean time in the on-state, and 5) mean time on the off-state. Using Bayesian inference, the model updates its prior probability distributions to posterior probability distributions, as shown in Equation (1). It does so by extracting and integrating information from the observed patient-specific data [50].

$$P(\Theta, H_t, B_t | D) = \frac{P(D, H_t, B_t | \Theta) \cdot P(\Theta)}{P(D)}, \quad (1)$$

The model's framework utilizes standard Bayesian inference, where $P(\Theta)$ is the prior parameter distribution, $P(D, H_t, B_t | \Theta) = P(D | \Theta, H_t, B_t) \cdot P(H_t, B_t)$ is the likelihood of the data given the parameters, and $P(D) = \int P(D, H_t, B_t | \Theta) \cdot P(\Theta)$ is the model evidence (observed data). The parameters are $\Theta = (\tau_{ON}, \tau_{OFF}, k, d, f)$ and latent variable (H_t, B_t) dynamics from LH profiling data. τ_{ON} and τ_{OFF} represent the probabilities that the value of pulsatile hypothalamic signal H_t will flip/remain the same; B_t represents

basal hypothalamic signal; d denotes clearance rate; k denotes maximum secretion rate, and f represents relative strength of two hypothalamic signals (Table 2).

Parameter	Description
τ_{ON}	Probability the pulsatile signal H_t will flip
τ_{OFF}	Probability the pulsatile signal H_t will remain the same
k	Maximum secretion rate
d	Clearance rate
f	Relative strength of two hypothalamic signals
H_t	Pulsatile hypothalamic signal
B_t	Basal hypothalamic signal

Table 2: Model parameters and their descriptions

Pulses are identified according to the expected value of the pulsatile hypothalamic signal, which ranges from 0 to 1. Subsequently, when a signal crosses the 0.5 threshold, the model indicates that a pulse is likely (Table 3). For example, if H_t —the pulsatile hypothalamic signal—was equal to 0.6 at 10 minutes and 0.2 at 20 minutes, we would expect a pulse only at the 10 minute mark (Table 4).

Table 3: Transition probabilities for the pulsatile hypothalamic signal H_t

H_t	$H_{t+\delta t}$	
	0	1
0	$1 - \frac{1}{\tau_{\text{ON}}} \cdot \delta t$	$\frac{1}{\tau_{\text{ON}}} \cdot \delta t$
1	$\frac{1}{\tau_{\text{OFF}}} \cdot \delta t$	$1 - \frac{1}{\tau_{\text{OFF}}} \cdot \delta t$

3.2.1 Advantages

Rather than a point-estimate, hormoneBayes provides quantified estimates of uncertainty using a Bayesian framework. This can also be advantageous as a clinical tool for physicians, where it can guide—rather than provide one-dimensional information to—their decision making [50]. Another advantage of a Bayesian approach is its flexibility in incorporating different types of data and modeling structures. For example, in postmenopausal women, parameter priors can easily be adjusted to reflect higher LH secretion rate. Indeed, hormoneBayes was extremely effective in predicting pulsatile dynamics across diverse groups. Consistent across five cohorts (healthy men, healthy premenopausal women, postmenopausal women, women with PCOs and women with hyperandrogenism), the model identified 89% of pulses [50]. Further, being a stochastic model, it is able to account for randomness. Hahl et. al point out that this may be advantageous especially in biological systems, where noise is inherent [22].

Table 4: Pulse identification based on expected hypothalamic signal H_t

Time (min)	Expected H_t	Pulse Detected?
0	0.12	No
10	0.60	Yes
20	0.20	No
25	0.48	No
30	0.52	Yes
35	0.71	No
50	0.55	Yes

3.2.2 Challenges

However, this approach is limited in that GnRH pulsatility is linked to a multitude of conditions. While an increased LH:FSH ratio is associated with PCOS, it is also linked to autoimmune diseases [20]. Moreover, this approach requires the precise measurement of a prior distribution to be accurate, which may be difficult to obtain. The model also assumes a linear clearance rate for LH circulation levels; Voliotis et. al anticipate that such an assumption should have minimal impact in comparison to multiple timescale clearance dynamics.

3.3 Modified Least-Squares & Monte Carlo Markov Chain

It is important to note that, although GnRH has a vital role in regulating LH/FSH ratio, it is unethical to measure in humans. For starters, it's present in extremely low concentrations in the bloodstream, has a short half-life, and rapidly fluctuates [13]. Therefore, its pulsatility is instead measured as a series of sharp spikes known as "Dirac impulses." hormoneBayes, although not computationally intensive, requires intensive prior patient-specific data.

Runvik et. al propose an alternative: estimating impulse timings, weights (pulse sizes), and elimination rates of GnRH. Since LH is much easier to measure in humans, their model's parameters are simply temporal LH levels. Their approach utilizes the least-squares method, which takes multiple "steps"—adjusting the timing and strength of GnRH spikes—to find the best match between their model and real-world data [40].

Reconciling Runvik et. al and hormoneBayes to provide a more robust and accessible diagnosis of LH pulsatility is a promising possibility. Both models operate on the same assumptions, such as pulsatile/basal release, a Bayesian statistical framework to infer latent hormone dynamics from noisy data, and linear clearance rate. Their main difference lies in the methods of handling measurement error. hormoneBayes incorporates measurement error as an additional parameter, η_t , that is determined based on the assay coefficient of variation. η_t is a normally distributed random variable with a mean of zero and standard deviation equal to the assay coefficient of variation (Equation 2). In contrast, Runvik et al. rely on underlying biochemistry in the form of

b_1/b_2 (time constants of GnRH/LH concentration eliminations) pairs. They assert an understanding that the true parameter values will satisfy the following inequalities: $0.23 \text{ min}^{-1} \leq b_1 \leq 0.69 \text{ min}^{-1}$, $0.0087 \text{ min}^{-1} \leq b_2 \leq 0.014 \text{ min}^{-1}$.

$$LH_t^{\text{obs}} = LH_t(1 + \eta_t) \quad (2)$$

4 Insulin Modeling

4.1 Short-Term Dynamics of Insulin Action on Glucose

Insulin resistance is one of the key features in patients with PCOS. When clinically treating such patients, it is important to note the differences in insulin action compared to healthy patients. Additionally, glucose levels fluctuate in the short-term. Factors such as digestion, absorption, and hormonal dysregulation can heavily impact plasma glucose concentrations. Hampton et. al's study investigates these relationships, introducing a model that captures the short-term dynamics of insulin acting on glucose in a population of adolescent girls with obesity and with or without PCOS. In this study, subjects consume an oral glucose tolerance test (OGTT), a physiological test where participants ingest glucose in the form of a sugary drink [23].

The Oral Minimal Model (OMM) is a differential-equations based model of glucose-insulin dynamics, and quantifies insulin sensitivity (S_1). There are two equations involved in the OMM:

$$\dot{G} = -[S_G + X_G]G + S_G G_b + \frac{Ra_{meal}}{V} \quad (3)$$

$$\dot{X}_G = \begin{cases} -p_2^G X_G, & I(t) < I_b \\ -p_2^G X_G + p_3(I(t) - I_b), & I(t) \geq I_b \end{cases} \quad (4)$$

$G(t)$ represents glucose concentration in mg/dL; G_b and I_b represent basal glucose and insulin concentrations; $X_G(t)$ represents insulin action on glucose; S_G represents glucose effectiveness; p_2^G is a time constant of insulin action; p_3 is a constant of insulin action clearance/appearance; and $Ra_{meal}(\alpha, t)$ is a piecewise-linear function that captures the presence of exogenous glucose in the bloodstream (Table 5). To solve these equations, initial values are defined as $G(0) = G_b$ and $X_G(0) = 0$ [7].

Fitting OGTT data to the OMM, findings suggest that the dynamics of insulin action on glucose are delayed. In non-obese youth, insulin peaks at 30 minutes post-drink, whereas adolescents with obesity and PCOS peak in insulin at 84 ± 47 minutes post-drink. Moreover, healthy teens have lower peak insulin concentrations, averaging $55 \mu\text{U/mL}$ as opposed to $361 \mu\text{U/mL}$ where U represents micro-units of insulin per milliliter of blood [23]. Through observing the metabolic features of patients, clinics can classify them as insulin resistant/non-resistant—that is, they can detect one of the key symptoms of PCOS.

The scope of the short-term dynamics study mentioned above is limited for two main reasons: 1) the age of subjects is within a constrained range, and 2) the diagnosis of PCOS is defined by only phenotype B of the Rotterdam criteria. Since this study only concerns adolescent girls, more research is to be needed regarding

Variable	Description
$G(t)$	Glucose concentration in blood (mg/dL)
G_b	Basal (fasting) glucose concentration (mg/dL)
$I(t)$	Plasma insulin concentration (μ U/mL)
I_b	Basal insulin concentration (μ U/mL)
$X_G(t)$	Insulin action on glucose (arbitrary units)
S_G	Glucose effectiveness (1/min)
p_2^G	Time constant of insulin action (1/min)
p_3	Insulin action rate parameter (1/min per μ U/mL)
$Ra_{meal}(\alpha, t)$	Rate of appearance of glucose from the meal (mg/dL/min)
V	Distribution volume for glucose (dL)

Table 5: Description of variables in the Oral Minimal Model (OMM)

insulin action in more mature populations, such as postmenopausal PCOS patients. Further, the study defines patients with PCOS as exhibiting hyperandrogenism and an irregular menstrual cycle, and is thus uninformed about insulin’s relationship with polycystic ovaries. This modeling complexity is furthered as insulin resistance is tissue specific, meaning it may be present at varying levels in different tissues. Arner and Rydén hypothesize that the development of insulin resistance in adipose tissue specifically may result in excess free fatty acids, which induces insulin resistance in other tissues [4].

4.2 Homeostasis Model Assessment

There are two widely used measures of insulin resistance: 1) Homeostasis Model Assessment (HOMA), and 2) hyperinsulinemic euglycemic clamps. Hyperinsulinemic euglycemic clamps first deliver a fixed amount of insulin into the bloodstream, creating a “hyperinsulinemic” state. Glucose is simultaneously delivered, but is not administered at a fixed rate. Instead, the rate is adjusted to maintain a normal blood sugar level. Researchers can subsequently measure a patient’s insulin sensitivity. Though hyperinsulinemic euglycemic clamps are referred to as the “gold standard” of determining insulin resistance, its methodologies are intensive and often inaccessible [5]. HOMA-IR, first developed in 1985, can be determined simply through fasting blood insulin and glucose levels. The formula for HOMA1-IR, a closed-form equation, is:

$$\text{HOMA-IR} = \frac{\text{Fasting insulin } (\mu\text{U/mL}) \times \text{Fasting glucose (mg/dL)}}{405} \quad (5)$$

This formula was derived from a population of healthy individuals, and subsequently determines a fixed constant of 405 [29].

Example:

As an example of utilizing HOMA1-IR, suppose that patient has a fasting insulin level of 18 μ U/mL and a fasting glucose level of 95 mg/dL. We can calculate their HOMA-IR as follows:

$$\text{HOMA-IR} = \frac{18 \times 95}{405} = \frac{1710}{405} \approx 4.22$$

Interpretation:

A HOMA-IR value above approximately 2.0–2.5 may suggest insulin resistance, depending on the population and lab standards. In this example, a value of 4.22 indicates elevated insulin resistance and likely warrants further clinical evaluation.

Meanwhile, HOMA2-IR is more computationally expensive to account for more variables as well as a non-linear relationship between insulin and glucose. Rather than a manual formula, HOMA2-IR is a computer-calculated model [28]. The performance of HOMA1-IR versus HOMA2-IR varies with population—in Type 2 diabetic patients, HOMA2-IR was more predictive than HOMA1 [45]. In another study, Safar et. al show that HOMA1-IR is consistently more associated with metabolic syndrome (MS) and PCOS, with an odds ratio of 5.7 versus 4.2 and 3.7 versus 3.5 [42]. This could point towards revision with computer-based methods; HOMA2-IR is intended to be more robust but underperforms in extreme cases of dysregulation.

Lastly, it is of note that insulin resistance’s correlation with PCOS is limited. Indeed, PCOS with insulin resistance were more clinically impacted. Yet, at a prevalence of 64%, insulin resistance does not appear to be a universal feature [15].

4.3 Multivariate Regression

Using stepwise regression analysis, Gennarelli et. al determine significant predictors for IR in PCOS. Pairwise models were evaluated to minimize prediction error, yielding three best prediction models. Model I uses waist circumference and fasting insulin (both with a p-value of < 0.001), and has an R-squared value of 0.77. Model II uses waist circumference and triglycerides (both with a p-value of < 0.001), and has an R-squared value of 0.65. Model III uses waist circumference and subscapularis skin fold (both with a p-value of < 0.01), and has an R-squared value of 0.64 (Table 6). Interestingly, despite waist circumference consistently being in the highest-performing models, androgen excess was not present. Gennarelli et. al propose, then, that androgen excess is only indirectly related to IR [18]. However, Unluhizarci et. al conclude that androgen excess per se contributes to IR. Antiandrogen therapy was even displayed to partially reverse peripheral IR resistance [48]. These conflicting results may be caused by the bidirectional relationships between insulin and androgens.

Model	Predictors (p-value)	R ²
Model I	Waist circumference, Fasting insulin ($p < 0.001$)	0.77
Model II	Waist circumference, Triglycerides ($p < 0.001$)	0.65
Model III	Waist circumference, Subscapularis skin fold ($p < 0.01$)	0.64

Table 6: Top predictive models for insulin resistance in PCOS from stepwise regression analysis [18]

5 Systemic Modeling

In addition to focusing solely on individual hormones, it is also important to consider PCOS through the lens of systemic modeling. Such modeling incorporates regulation through both the HPG axis and by mechanisms of ovarian testosterone production.

Hendrix et. al model pituitary feedback and inter-ovarian follicular growth properties, with a primary focus on these systems as functions of circulating total testosterone levels. Building upon previous literature, their parameters also include LH, FSH, estradiol, progesterone, Inhibin A, and Inhibin B. Although the study also models inter-ovarian follicular growth properties, this review will focus on their modeling of pituitary feedback. Ultimately, Hendrix et. al propose 16 ordinary differential equations. These equations are clinically relevant; they have stable periodic solutions indicating a sustained and plausible hormonal state—whether that be healthy or dysfunctional. One important underlying assumption behind these equations is that baseline synthesis rate of LH and FSH are assumed to be dependent on testosterone [24]. Equations (6) and (7) represent LH and FSH models, respectively, with initial value vectors determined from patients on the first day of the study:

$$\frac{d}{dt}LH = \frac{1}{v} \cdot k_{LH} \cdot \left(\frac{1 + c_{LHP} \cdot P4^6}{1 + c_{LHe} \cdot E2^5} \right) \cdot RP_{LH} - r_{LH} \cdot LH \quad (6)$$

$$\frac{d}{dt}FSH = \frac{1}{v} \cdot k_{FSH} \cdot \left(\frac{1 + c_{FSHP} \cdot P4}{1 + c_{FSHe} \cdot E2^5} \right) \cdot RP_{FSH} - r_{FSH} \cdot FSH \quad (7)$$

Here, v represents blood volume; k_{LH} represents decay rate, c_{LHP} represents the coefficient for progesterone's impact on LH (negative feedback), c_{LHe} represents the coefficient for estradiol's impact on LH (positive feedback), RP_{LH} represents the LH reserve pool (the amount of LH stored in the pituitary and available for release), and r_{LH} represents the basal LH secretion rate (Table 7). As exhibited by Equation (7), FSH is modeled in a similar manner.

$$\begin{aligned} \frac{d}{dt}RP_{LH} = & \frac{v_0 \cdot T(t - d_T)^r + v_1 \cdot \left(\frac{E2(t - d_{E2})^r}{(KM_{LH}^a + E2(t - d_{E2})^r)} \right)}{1 + \left(\frac{P4(t - d_{P4})}{K_{ILH}^r} \right)} \\ & - k_{LH} \cdot \left(\frac{1 + c_{LHP} \cdot P4^6}{1 + c_{LHe} \cdot E2^5} \right) \cdot RP_{LH} \end{aligned} \quad (8)$$

$$\begin{aligned} \frac{d}{dt}RP_{FSH} = & \frac{V_{FSH}}{1 + \left(\frac{InhA(t - d_{InhA})}{K_{InhA}} \right) + \left(\frac{InhB(t - d_{InhB})}{K_{InhB}} \right)} \\ & - k_{FSH} \cdot \left(\frac{1 + c_{FSHP} \cdot P4}{1 + c_{FSHe} \cdot E2^5} \right) \cdot RP_{FSH} \end{aligned} \quad (9)$$

In the reserve pool equations, the production and decay rate variables are: v_0 , denoting basal testosterone production rate; v_1 , denoting estradiol production rate; d_{rp} , denoting testosterone decay rate (and similarly, for estradiol, InhA, and InhB); KM_{LH}^a , denoting the Michaelis constant for LH activation effect; K_{ILH}^r , denoting the inhibition constant; and V_{FSH} , denoting the maximum rate of FSH production (Table 8). Equations (8) and (9) are similarly adapted for LH and FSH reserve pools. The differences, to account for FSH, lie in the

	LH	FSH
Variable	LH (IU/L)	FSH (IU/L)
Rate of Change	$\frac{d}{dt} LH$	$\frac{d}{dt} FSH$
Volume	v (L)	v (L)
Decay Constant	k_{LH} (1/time)	k_{FSH} (1/time)
Progesterone Feedback	c_{LHP}	c_{FSHP}
Estradiol Feedback	c_{LHe}	c_{FSHe}
Current Hormone Level	LH	FSH
Reserve Pool	RP_{LH} (IU)	RP_{FSH} (IU)
Basal Secretion Rate	r_{LH} (1/time)	r_{FSH} (1/time)
Other Hormones	$P4, E2$	$P4, E2$

Table 7: Variables and parameters in the LH and FSH rate-of-change equations

synthesis terms (v_0 , v_1 , and V_{FSH}). The synthesis terms now have constant growth and inhibition due to $InhA$ and $InhB$ (Table 9).

	RP_{LH}	RP_{FSH}
Variable	RP_{LH} (IU)	RP_{FSH} (IU)
Rate of Change	$\frac{d}{dt} RP_{LH}$	$\frac{d}{dt} RP_{FSH}$
Max Synthesis Rate	v_0, v_1	V_{FSH}
Estradiol Effect	$E2(t - d_{E2})$	$E2$
Estradiol Constant	$K M_{LH}^a$	c_{FSHe}
Delays	d_T, d_{E2}	d_{InhA}, d_{InhB}
Inhibin Feedback	—	$InhA, InhB, K_{InhA}, K_{InhB}$
Time Delay on T	$T(t - d_T)$	—
Hill Coefficient	r	—

Table 8: Variables and parameters in the reserve pool equations for LH and FSH

Additionally, discrete time-delays are accounted for in the synthesis terms. Discrete time-delays represent the time between a hormonal signal and its feedback.

Example:

	LH	FSH
Decay Term	$k_{LH} \cdot \left(\frac{1+c_{LHP} \cdot P4^6}{1+c_{LHe} \cdot E2^5} \right) \cdot RP_{LH}$	$k_{FSH} \cdot \left(\frac{1+c_{FSHP} \cdot P4}{1+c_{FSHe} \cdot E2^5} \right) \cdot RP_{FSH}$
Reserve Pool Synthesis	$\frac{v_0 T(t - d_T)^r + v_1 \cdot E2(t - d_{E2})^r}{(KM_{LH}^a + E2(t - d_{E2}))^r}$	$\frac{V_{FSH}}{1 + \frac{InhA(t - d_{InhA})}{K_{InhA}} + \frac{InhB(t - d_{InhB})}{K_{InhB}}}$
Estradiol Feedback	Positive (via $E2$ denominator)	Positive (via $E2$ denominator)
Inhibins (InhA/B)	Not included	Inhibitory effect in synthesis
Delay Terms	d_T, d_{E2}, d_{P4}	d_{InhA}, d_{InhB}
Activation/Inhibition	Michaelis-Menten via KM_{LH}^a	Inhibition via K_{InhA}, K_{InhB}

Table 9: Comparison of structural features in LH and FSH modeling equations

As an example of utilizing Equation 6, suppose we have the hypothetical values: $v = 5$, $k_{LH} = 0.3$, $c_{LHP} = 0.1$, $c_{LHe} = 0.05$, $P4 = 2.0$, $E2 = 1.5$, $RP_{LH} = 10$, $r_{LH} = 0.05$, and $LH = 1.2$.

Then, using the equation to compute the rate of change of LH with respect to time:

$$\begin{aligned}
\frac{d}{dt} LH &= \frac{1}{5} \cdot 0.3 \cdot \left(\frac{1 + 0.1 \cdot 2.0^6}{1 + 0.05 \cdot 1.5^5} \right) \cdot 10 - 0.05 \cdot 1.2 \\
&= 0.06 \cdot \left(\frac{1 + 0.1 \cdot 64}{1 + 0.05 \cdot 7.59375} \right) \cdot 10 - 0.06 \\
&= 0.06 \cdot \left(\frac{1 + 6.4}{1 + 0.3797} \right) \cdot 10 - 0.06 \\
&= 0.06 \cdot \left(\frac{7.4}{1.3797} \right) \cdot 10 - 0.06 \\
&= 0.06 \cdot 5.364 \cdot 10 - 0.06 \\
&= 3.2184 - 0.06 \\
&= \boxed{3.1584}
\end{aligned}$$

Interpretation:

Therefore, the rate of change of LH with respect to time in this scenario is approximately 3.1584 IU/L (International Units per Liter).

When conducting systematic modeling, it is imperative to evaluate the interactions between hormones. Prior research reveals that insulin-mediated testosterone production in the ovaries is a key trait in PCOS patients. Insulin binds to its receptors on ovarian thecal cells (which produce testosterone), then use the signaling molecule inositolglycan to trigger the upregulation of testosterone production [34]. Thus, Graham and Selgrade determine a physiological role for both testosterone and hyperinsulinemia in the ovulatory cycle. They do so by

adapting and modifying the original sixteen proposed differential equations from Hendrix et. al. According to their semi-mechanistic model, insulin-mediated testosterone production is positively correlated with ovulatory disruption. Instead of focusing on the origin of hyperinsulinemia, their model suggests a role for mechanisms behind insulin-mediated testosterone production. In addition, Graham and Selgrade reveal that insulin acts as a co-gonadotropin with LH, thereby mediating LH sensitivity [21].

6 Insights from Machine Learning

6.1 Relevance of Machine Learning in Computational Modeling

Machine learning models have the ability to capture complex patterns, which can be neglected in simplified mathematical models that are based on mechanistic governing equations [6]. However, they are limited when it comes to black-box models, which, if only trained on experimental data, are uninformed of the biological mechanisms. Nevertheless, there are still insights to be gained from machine learning. Particularly, it is effective in quantitatively assessing key factors involved in modeling. For instance, Elmanai uses both tree-based models and recursive feature elimination (RFE) to rank importance of selected features. Tree-based models, hence their name, make conclusions using decision trees. They consider the data at each node, and take a different "path" to a node depending on the comparison being made at the initial node. RFE is the process of finding the most important features. It is performed by first training a model on all available parameters, then analyzing the model coefficients to remove non-significant features. Consistent across both models, age, weight, and height are significant predictors for PCOS [17].

7 Standard Machine Learning Models

In one study, Barrera analyzes 135 machine learning models that predict PCOS diagnosis. Of these models, common algorithms included support vector machines (SVM), K-nearest neighbors (KNN), and random forests (RF) [17]. In this section, I discuss the differing outcomes and suitability of three selected models for representing PCOS dynamics. While numerous computational models exist, this brief review of models will focus on a representative subset rather than providing an exhaustive analysis.

7.0.1 Support vector machines (SVM)

Support vector machines are typically used for classification tasks, and operate by finding the optimal hyperplane to separate given data points. They're also advantageous as they have a lower prediction error in comparison to artificial neural networks. When being used to identify organic molecules that modulate the function of G-protein coupled receptors, for instance, SVM models performed considerably better than artificial neural networks [9]. This result is exemplified when a large number of features are being considered, making it appropriate for the heterogenous symptoms of PCOS.

7.0.2 K-nearest neighbors (KNN)

K-nearest neighbors is flexible in that it can be used for both classification and regression. The model first chooses k neighbors to evaluate, though this number can be tuned depending on sensitivity to noise. However,

KNNs are sensitive to outliers, which are prevalent in patients with PCOS. Moreover, in choosing an arbitrary k value, there exists a trade-off: larger k -values may decrease sensitivity, but we risk losing information on more complex relationships. Conversely, smaller k -values may increase sensitivity, but we are able to capture more complex relationships. Though Leslie et. al achieve an accuracy of 100% with a training ratio of 90:10 and $k = 11$, it should be acknowledged that PCOS datasets are prone to overfitting. Still, the complexity of the relationship captured is to be determined. A model that captures complex relationships in the training data is prone to overfitting, meaning it won't perform well in real-world scenarios.

7.0.3 Random forests (RF)

Similar to KNNs, random forests are used for both classification and regression tasks. First, they build multiple decision trees, with each tree being trained on a random subset of the data. At each “branch” of the tree, a random subset of features are chosen to make a prediction. Classification tasks rely on the popular decision of each tree, whereas regression tasks take the average prediction of each tree.

7.1 Results

Despite the focus on PCOS as a primarily hypoandrogenic disorder, the Rotterdam criteria also outlines phenotype D: patients with polycystic ovaries and anovulation. Wang et. al analyze the difference in steroid hormones in PCOS patients with hyperandrogenemia (HA) and non-HA, as well as non-PCOS patients. They train eight machine learning models based on levels of serum androgens, corticosteroids, progestins, and estrogens. A random forest model is ultimately most successful, with diagnostic sensitivity of 100%, 93.3%, and 91.7% for HA PCOS, NA PCOS, and non-PCOS patients, respectively. These findings underscore the potential usage of steroid hormone profiling in aiding clinical diagnosis, especially given the heterogeneous clinical and hormonal presentations among PCOS patients [51].

7.2 Limitations to Machine Learning

Several machine learning models—including clustering algorithms, neural networks, and deep learning—capture interactions between predictors. As a result, dependencies within the input data parameters are assumed, which deem such models unreliable for outliers. This unreliability can be problematic, especially for atypical manifestations of PCOS. For instance, although obesity is a standard phenotypic feature of PCOS, it is not consistent across all patients with PCOS. Previous studies about lean PCOS patients indicate that insulin resistance is independent of obesity [47]. Thus, the models mentioned above, which place large significant scores on predictors like BMI, can be inefficient in lean PCOS patients.

7.3 Optimizing Machine Learning Models

Baseline models presented in this review thus far have exhibited relatively high accuracy, with most achieving 90%. Future work in machine learning for PCOS diagnosis could continue improving accuracy with more advanced techniques. Namely, recent work from Elmannai et. al proposes the stacking of well-established, high-performing baseline models with a meta-learner—that is, outputs from baseline models are pooled as

inputs to a meta-model. Additional Bayesian optimization and REF-based feature selection was employed on the stacked model, achieving an accuracy of 100% [17].

Due to the high precision required as inputs to physiological, deterministic models, high dependence on hormone levels can be impractical in a clinical setting. Instead, work from Shanmugavadivel et. al and Moral et. al reflects inclusion of clinical data [32, 43]. Shanmugavadivel et. al also include ultrasound imaging data in addition to clinical features (testosterone, family history, BMI, etc.), and achieve an accuracy of 98.29% using a VGG16 transfer learning algorithm [43]. Moral et. al outline a classification similar to that of Shanmugavadivel et. al; however, rather than optimization of a meta-model, they optimize feature extraction from ultrasound images using a technique known as CystNet. They achieve an accuracy of 97.75%, which is comparable to that of Shanmugavadivel et. al [32].

8 Future Computational Model Considerations

As determined by previously mentioned machine learning models, multimodal data is highly effective in PCOS prediction. This idea is not only supported through machine learning, but through physiological studies as well. Insulin resistance is linked to increased serine phosphorylation, which modulates P450c17, a key enzyme in androgen production. Thus, it is likely that a single defect produces both insulin resistance and hyperandrogenism in PCOS [16]. In addition, there is a strong correlation between testosterone levels and insulin resistance: women with high testosterone are 2.78 times more likely to exhibit insulin resistance [27]. Potential areas for future computational models are incorporating such multimodal data, such as comparing the performances of insulin-based versus LH-based models, or incorporating the underlying mechanism that drives both IR and hyperandrogenism (rather than relying on a bidirectional relationship between the two hormones).

9 Concluding Remarks

When evaluating models, it is of note to consider Cobelli's guidelines for endocrine-metabolic systems. Namely, a model should have a degree of identifiability in both *a priori* and *a posteriori*. *A priori* refers to theoretical parametric determination—that is, if all parameters were to be recorded namelessly, an identifiable model should still have unique parameters. On the other hand, *a posteriori* refers to practical significance—i.e. if the model can tolerate noise in its parameters [12]. Indeed, each model discussed exhibits identifiability in accounting for measurement error, a systemic perspective, interactions in hormonal loops, etc.

Due to the multifaceted nature of its etiology, current treatments for PCOS focus on managing endocrinological symptoms. Patients often use oral contraceptives—most commonly estrogen and progesterone combinations—to lower testosterone levels. Lowering testosterone levels can alleviate symptoms such as hirsutism and acne [14]. Emerging therapies also include androgen modulators, incretin modulators, insulin sensitizers, and neuroendocrine modulators [19]. Systematic modeling of the HPG axis may be of significance here; relationships between testosterone and LH/FSH can be investigated prior to clinical testing of oral contraceptives.

Lastly, I consider the integration of computational models with current diagnostic tools. Clinicians diagnose based on the Rotterdam criteria, which are analyzed through three main methods: 1) clinical symptoms, 2) laboratory evaluations of biochemical hormonal profiles, and 3) ultrasound imaging for presence of polycystic ovary morphology. High dependence on clinical symptoms is unfavorable; instead, we can turn to hormone lab work as a more accessible diagnostic tool than ultrasound imaging. The deterministic models outlined in this review can be well-integrated into clinical settings. This is because all of the hormones’—LH, FSH, insulin, and testosterone—levels are most commonly determined through blood serum tests. Though GnRH pulsatility is difficult to determine, estimations can be provided through LH/FSH models. Moreover, insulin models such as the OMM determine metabolic features of patients. Less computationally/resource-expensive models of HOMA or multivariate regression can also provide benchmarks for insulin resistance, but point towards irregularities in computer programs and potential bidirectional relationships with androgens. Recent work in such a relationship is conflicting, though can be explored further through systematic modeling. As shown by Graham and Selgrade, modeling LH/FSH as a direct function of testosterone is indeed successful [21]. There has also been evidence towards insulin-mediated androgen production—that is, a bidirectional relationship between insulin and androgens—despite prior conclusions from research in men disagreeing [36].

Due to the partially unknown interactions of underlying mechanisms behind hormone interactions, machine learning provides a promising approach. Thus, using both mechanistic models in addition to information that can be processed via machine learning will reach a broader range of PCOS patients. Information that can be processed via machine learning could include ultrasound imaging with neural networks, or clinical patient metadata. Hybrid approaches could capture the full spectrum of the Rotterdam criteria; patients can be diagnosed both hormonally and morphologically.

Overall, both mechanistic and data-driven approaches from ML enhance clinical diagnostic power. Modeling also provides a strong framework for emerging questions—such as the relationship between insulin and androgens, the mechanisms behind prevalent IR in PCOs patients, and more. Future refinements of these models may reveal new pathways to target and help personalize treatments for the heterogeneity of hormonal profiles in PCOS patients.

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